PROJECT REPORT

Project Similarity Detection System

An explainable academic project comparison prototype for faculty-led review.

Prepared for	Demonstration date	Run mode
Faculty project demonstration	24 October 2026	Local Windows prototype

Project purpose

The system compares a submitted academic project with historical project records and presents ranked candidates with visible evidence. Faculty can inspect scores, shared terms, and explanations before making an academic assessment. Similarity is a review signal, not an automatic plagiarism finding.

Work present in the project

- **Web interface:** React and Vite pages for submission, repository browsing and filters, project details, method explanation, and results.

- **Backend service:** FastAPI routes for project records, health checks, algorithms, and analysis submission and retrieval.

- **Local data store:** SQLAlchemy models with SQLite by default and ten seeded demonstration records on an empty database.

- **Similarity engine:** TF-IDF/cosine text comparison, Jaccard term overlap, MinHash over three-token shingles, and normalized source-token comparison. Optional local Ollama embeddings have a TF-IDF fallback.

- **Evidence and review:** Weighted scores, ranked candidates, shared terms, explanations, side-by-side comparison, print action, and faculty decision/notes interface.

- **File handling:** PDF/TXT reports and supported source files/ZIPs. Uploaded source is read as text and never executed.

Scoring model

Each available component contributes according to its base weight. Missing report or source evidence is omitted and the remaining weights are renormalized.

Evidence component	Method	Weight	Demo result
Title language	TF-IDF + cosine	10%	75%
Abstract and description	TF-IDF + cosine	25%	28%
Project report	TF-IDF + cosine	20%	27%
Declared keywords	Jaccard overlap	15%	31%
Text shingles	MinHash signatures	10%	2%
Source-code structure	Normalized code + cosine/Jaccard	20%	Unavailable

DEMONSTRATION WALKTHROUGH

What the live run showed

A sample submission titled "Project Similarity Detection System" was compared with all ten seeded repository candidates. The closest candidate was "Project Similarity Detection."

31%	MODERATE SIMILARITY

How to present the result

1. Explain the input. The submission includes a title, project description, implementation notes, keywords, academic year, report text, and a source file.

2. Show the repository. The first launch seeds ten demo records when the local projects table is empty. They are examples, not an institutional archive.

3. Read the ranking. The closest candidate appears first, followed by other candidates in descending score order.

4. Inspect evidence. Review component scores, shared terms, and explanations. Missing evidence is labeled unavailable, not zero.

5. Leave the decision to faculty. Scores prioritize review; they do not establish copying, intent, misconduct, or originality.

Local demonstration setup

From the project folder on Windows, run `run-website.ps1`. The site opens at `http://127.0.0.1:5173`. Keep both services running during the presentation. First-time setup instructions are in `FACULTY_DEMO_GUIDE.md`.

Scope and limitations

This is a local prototype with demo data. Production use needs an approved institutional archive, authentication, access controls, and persistent reviewer records. Faculty notes are saved only in the active browser session. The source-code score is unavailable here because seeded candidates do not include source code. Local Ollama embeddings are optional; the live run used TF-IDF.

Faculty conclusion: use the displayed match as a prompt for contextual review. Do not interpret the score alone as an academic misconduct decision.